

Verified Cleaning Plans: Plan-Level Selective Prediction Turns Local LLM Planners into Trustworthy Table Cleaners

Ricardo Alanis
ricardo.alanis@gmail.com

June 2026

Abstract

Cleaning messy tabular data—particularly *canonicalization*, the merging of inconsistent surface forms such as USA/U.S.A/united states into one canonical value—resists rule-based automation and is routinely done by hand. We present ScrubData, an architecture built around a trust contract. A local LLM *planner* reads an aggregated column profile (per-value frequency counts, invariant to row count) and *proposes* a structured JSON cleaning plan; a deterministic executor *applies* it, making every change auditable and reversible; and *plan-level selective prediction* — a deterministic verifier that scores every proposed mapping and drops low-confidence entries to review flags — extends abstention from cell-level confidence to the plan itself. The verified union of the gated model plan with a reference-grounded heuristic is the architecture’s operating point: a zero-configuration, zero-label system that repairs 41% of the hospital benchmark’s 509 real errors at 0.905 precision (strongest of three training seeds; 3-seed mean 0.891 ± 0.012 at 0.396 ± 0.025 coverage, $\pm = 95\%$ CI), with every declined merge surfaced for review. Four deterministic capabilities — profile-level *suspect surfacing* for high-cardinality columns, reconciliation against a pluggable *entity reference* built from open vocabularies, *cross-row majority voting* over repeated-entity groups, and *convention-conservatism* gates — carry the system to never-seen tables: macro F1 0.363 at 0.0219 damage over the 35 unseen-source pairs of a 42-pair benchmark, and *zero silent edits* across 35 wild tables plus a 239-table trust audit, released together as the WILDCLEAN benchmark.

Finally, we report where the capability lives. *Execution-verified* synthetic supervision — a training example is kept only if executing its plan provably recovers the known-clean table — buys the 4B fine-tune real in-distribution skill and the most precise gated planner at usable coverage (0.993 precision at 0.287 coverage); but five further retrains and a three-arm GRPO pilot leave held-out generalization statistically bounded (TOST against a pre-registered margin), while two of three zero-shot 24–31B open-weights planners (devstral-24B, gemma4-31B) dropped into the *identical* harness exceed the fine-tune’s operating point (0.915 precision at 0.485 coverage) with no task training. The architecture is planner-agnostic: it converts capability gains into trustworthy operating points without retraining. The shipped system runs entirely locally on commodity hardware; no data leaves the machine (the scaling-arm planners were measured via hosted endpoints; all are locally deployable open weights).

1 Introduction

A large share of practical data work is cleaning: a sales export where the same country is spelled four ways, a hospital roster where `birminghamx` should be `birmingham`, a CRM dump with mixed date formats and duplicated contacts. The fuzzy half of this work—recognizing that distinct surface forms denote the same entity—is exactly what rules do poorly and humans do slowly.

Large language models can do this fuzzy matching, but deploying them as cell editors has three problems. First, *trust*: a model that edits cells directly can silently corrupt data, and its errors are unauditable. Second, *cost and privacy*: shipping every row of a private table to a hosted frontier model is expensive and often unacceptable. Third, *hallucination*: asked for a canonical form, a generative model will invent one, and on tail entities it will invent wrong ones.

ScrubData addresses all three with an architecture in which the model never touches data. A profiler aggregates each column into a value-frequency distribution; a small local model reads the profile and *proposes* a JSON cleaning plan; a deterministic pandas executor *applies* it. The plan is the complete, inspectable, reversible specification of every change—there are no silent edits by construction (§3). Because the prompt scales with the number of *distinct* values rather than rows, a million-row table profiles like a hundred-row one.

This paper makes five contributions:

1. **A planner/executor decomposition with plan-level selective prediction:** the model proposes, a deterministic engine executes with full lineage, and a deterministic verifier gates every proposed mapping, extending abstention to the plan itself. The verified union of the gated model plan with a reference-grounded heuristic repairs 41% of hospital’s 509 real errors at 0.905 precision with zero configuration and zero labels (§3, §3.4, §5.1).
2. **WildClean and an un-gameable evaluation:** a 65-dataset suite (real-error benchmarks plus seeded error injection over 15 harvested open-data domains) scored with a churn-neutral, convention-tolerant metric that cannot be inflated by mass rewriting, with damage and silent edits scored alongside repair F1, degenerate baselines pinning the metric’s floor and ceiling, and the scorer itself validated against 30 adversarial known-by-construction cases (§4, §5.10).
3. **Four deterministic capabilities that carry never-seen-table generalization:** bounded suspect surfacing for high-cardinality columns, generic entity-reference reconciliation with an exact-hit typing floor, cross-row majority voting with a false-consensus guard, and convention-conservatism gates — each motivated by a measured failure regime and gated by the verifier (§3.5, §5.4).
4. **Execution-verified synthetic supervision,** the training method behind the 4B planner instantiation: every training example is validated by running the executor on the (dirty table, plan) pair and checking that the known-clean table is recovered; non-recovering examples are discarded (§3.2).
5. **A unified finding on where capability lives in this architecture:** five further supervised fine-tunes and a three-arm GRPO pilot with the executor as a verifiable reward leave held-out generalization statistically bounded (TOST), while two of three zero-shot 24–31B planners dropped into the same harness exceed the fine-tune’s operating point — deterministic machinery plus plan-level verification carry the generalization that exists, and raw planner capability, not task fine-tuning, scales it (§5.5, §5.6).

We deliberately report a negative-flavored finding alongside the positive ones: on *injected* typos, classical frequency clustering remains a strong baseline—by construction, injection places the canonical form in the column, which is clustering’s ideal regime. The advantage of grounding is concentrated where it matters: real errors, tail entities absent from the column, and adversarial near-misses where acting at all is wrong (§5).

2 Related Work

Error detection and repair. Raha and Baran [1] established configuration-free error detection and correction benchmarks (hospital, beers, flights, rayyan), which we adopt as out-of-distribution evaluation. HoloClean [2] combines integrity constraints, external reference data, and statistics in probabilistic repair, demonstrating that external signals can veto statistically plausible but wrong repairs—an insight our reference-veto inherits. GARF [3] learns repair rules self-supervised from the data itself; it also demonstrates the structural limit we observe for frequency-only methods: a lone categorical column offers no co-occurring signal to vote against an error.

The 2025–26 landscape. Post-Cocoon work concentrates on zero-label *detection*: ZeroED [7] (cloud-LLM cluster labeling, hospital detection F1 0.81, collapsing to 0.27 on smaller models), ForestED [8] (LLM-induced decision trees, 0.756), and Auto-Test [9] (corpus-mined semantic-domain constraints, no LLM at inference) — none performs zero-label *repair*. GIDCL [10] sets the labeled-class repair ceiling (hospital 0.97 with 20 labels and a LoRA trained per cleaned table); Cocoon [6] remains an unreproduced preprint (15 citing papers, none a reproduction). Two concurrent results corroborate facets of this paper’s central negative finding that machinery, not weights, carries cleaning generalization: a study showing even frontier models cannot correct table distortions without explicit priors [12], and a large multi-agent-debate evaluation in which LLM self-critique *degrades* repair and only an adversarially separate, execution-grounded critic helps [13] — the architecture our verifier instantiates. Spreadsheet-RL [11] reports the complementary positive case: with full-scale RL infrastructure and execution-verified rewards, a 4B model’s spreadsheet-manipulation skill *does* move (12.0% $\rightarrow$ 23.4%) — consistent with our reading that the gap between our \$30 pilot and such results is infrastructure scale, a boundary we state rather than blur (§5.5).

LLMs for data wrangling. Narayan et al. [4] showed frontier foundation models handle entity matching and imputation few-shot; Jellyfish [5] and Table-GPT [16] fine-tune mid-size models for data tasks. RetClean [17] is closest in spirit: retrieval from data lakes grounds cell repair, with the key empirical split that parametric knowledge suffices on world-known head values but collapses on the tail—motivating retrieval. Our work differs in the planner/executor decomposition (the model emits no cell values, only plans), in execution-verified supervision, and in the calibrated-abstention contract.

Entity linking over tables. TURL [18] and TableLlama [19] inject candidate entities into table understanding; Belotti et al. [20] show retriever coverage is the accuracy ceiling for table entity disambiguation and that long candidate lists hurt smaller models. RACOON [21] shows inference-time KG retrieval lifts a frozen model substantially, supporting our choice to ground at inference rather than bake aliases into weights (TURL’s out-of-domain collapse is the cautionary result). MTab [22] established type-constrained matching with abstention in semantic table annotation.

Clustering-based cleaning tools. The de-facto practitioner baseline is OpenRefine: key-collision (fingerprint) clustering plus a nearest-neighbour mode; we reimplement both faithfully, including blocking, and compare head-to-head.

Selective prediction. Risk-coverage analysis and calibration metrics [23] formalize “knowing when not to act”; to our knowledge their application to data-cleaning merge decisions is new.

Small specialized models. OpenMed [24] fine-tunes sub-500M encoders to state-of-the-art biomedical NER, the sister result to our thesis that small specialized models beat large generic ones on narrow structured tasks; we adopt their released PII token classifiers for column typing (§3.6).

3 Method

3.1 Planner / executor decomposition

A *profiler* reduces each column to a typed summary: detected semantic type, missing counts, issue flags, and a value–frequency distribution capped at 80 distinct values (high-cardinality columns are summarized by their head). The *planner*—either a deterministic heuristic or our fine-tuned 4B model—maps the profile (plus three sample rows) to a JSON plan: a list of per-column operations drawn from a closed vocabulary (`canonicalize_categories` with an explicit mapping, `parse_date`, `standardize_phone`, `mask_pii`, ...), table operations, and review flags. The *executor* applies the plan with pure pandas transforms. The plan is the only channel through which data changes: every diff is attributable to a named operation with a rationale, the original table is never mutated, and abstentions are first-class plan objects. We export per-run decision summaries as OpenTelemetry GenAI spans.

3.2 Execution-verified synthetic supervision

Training pairs are generated by corrupting clean synthetic tables with realistic noise (casing, aliases, single-character typos with Zipf-distributed long-tail categorical columns of 30–80 distinct values) while recording the ground-truth plan. The defining step is *verification by execution*: a candidate example is kept only if `EXECUTE(dirty, plan) = clean cell-for-cell`. This closes the loop between supervision and semantics—a plan that would not actually clean the table can never become a training label. We augment with real supervision derived from paired dirty/clean benchmarks by aligning cells and keeping only *learnable* canonicalizations (a surface form that is a string variant of its target and never a legitimate value elsewhere), which excludes unlearnable per-cell corrections such as divergent flight times. The fine-tune is QLoRA (rank 32) over Qwen3-4B-Instruct in bf16; one practical finding is that the base model’s tool-calling prior dominates free-running generation even after convergent fine-tuning (loss 0.16) and must be suppressed at decode time by banning the two tool-call tokens.

3.3 Reference-grounded canonicalization with abstention

For columns whose values reconcile to a known concept type (countries, administrative regions, cities), canonical forms are never generated: a fuzzy retriever (normalized edit similarity with first-character blocking and length prefilters) matches each distinct value against the type-scoped reference (ISO/pycountry; GeoNames cities500, 196k entries). A value maps to a canonical only if (i) similarity clears a threshold $\tau=0.84$, (ii) the best–second-best margin clears 0.03 (ambiguity veto: a value equally close to `Box` and `Boaz` abstains), and (iii) the canonical is cast to the column’s observed case convention. Near-misses ($0.70 \leq s < \tau$) are surfaced as review flags. The same wrapper grounds the *model* planner: for reference-typed columns the model’s free-generated mapping is replaced by the grounded one, so the model can add coverage but never invent a canonical for a grounded type.

3.4 Plan-level selective prediction: the verified union planner

Grounding constrains reference-typed columns, but the planner’s *free* canonicalization mappings on non-grounded columns remain unguarded—and they are where real-data precision dies (the fine-tune’s raw hospital plan: 0.185 precision at 0.475 recall). Rather than retrain, we extend abstention to the plan itself. A deterministic *verifier* scores every proposed mapping entry *raw*→*canon* with

contract-preserving evidence (no cell values emitted, no gold access): three hard gates distilled from the model’s measured failure classes—a value occurring ≥ 3 times is data, not a typo (*errors are rare*); the target must be a frequent column value clearly dominating the source (no mapping one typo onto another); digit-bearing codes repair only when the letter part is near-identical—then a confidence combining edit similarity with frequency support. Entries below a threshold τ are dropped to review flags; abstention stays first-class. Sweeping τ yields a plan-level precision–coverage curve. The shipped composition, the *verified union planner*, is the verifier-gated model plan ($\tau=0.5$) unioned with the grounded heuristic’s mappings (the model wins per surface form); the same code path is the product default.

3.5 Visibility and consensus: four deterministic capabilities

Four further mechanisms, each motivated by a measured failure regime on never-seen tables, complete the deterministic machinery. **(a) Suspect surfacing.** The profile’s value-frequency view is capped, so high-cardinality columns hide their dirty cells from any planner. Every column profile now carries a bounded `suspect_values` section: rare anomalous surfaces with evidence-backed repair candidates (frequency dominance, edit similarity, reference membership). The heuristic planner repairs from suspects under a strict verifier bar ($\tau_{hc}=0.8$) and flags the rest. **(b) Generic entity reference.** Open vocabularies (SemTab ToughTables aliases — derived excluding our benchmark tables; MusicBrainz search-hint misspellings; RxNorm; Wikidata; ROR) register as a pluggable reference type. Because the reference is broad, entity-typing a column additionally requires that $\geq 20\%$ of its distinct values match the reference *exactly* — fuzzy coverage alone overfires on name-like columns (measured). This resolves the regime where every surface in a column is unique (no in-column frequency signal exists at all): five such benchmark tables go from 0.0 to 0.955–0.957 F1 at *zero* damage. **(c) Cross-row majority voting.** Tables that repeat a real-world entity across rows (a flight reported by many sources) carry their own repair signal. A detection step finds compact-token key columns with small groups (median multiplicity 3–30) and columns whose groups show *majority-bearing* disagreement with per-group information; a table-level operation then resolves thin dissenting minorities to the group majority. A *false-consensus* guard declines when minority shares look like legitimate correlated updates rather than reporting errors (mean minority share ≥ 0.25) — a flat volume cap was measured to destroy the legitimate dense-disagreement regime and replaced. **(d) Convention conservatism.** The planner never re-formats an internally consistent column: date and percent ops are gated on dominant-shape inconsistency (digit and alpha runs collapsed; 90% rule), ZIP/postal-named columns are never typed as phones or dates, and Excel-serial date typing requires a date-suggestive column name. Suppressed minority values surface as review flags — abstention is visible, never silent. The verifier enforces the same gates on model-emitted plans at the verification boundary.

3.6 PII as a second task instance

The identical contract covers PII: a deterministic tier types columns by checksum and pattern validators (Luhn, IBAN mod-97, SSN/email/phone) over distinct values; an optional 44M OpenMed-PII token classifier [24] extends coverage to names and addresses, gated by a sensitive-type allowlist and a column-level coverage vote; and masking, salted hashing, and join-stable pseudonymization are deterministic executor operations. Measured briefly: the classifier, though trained on sentence-level clinical text, transfers to bare cell values — 100% detection on person-name cells and 100% on address cells ($n=40$ sampled cells each); the validator tier, evaluated out-of-distribution on per-type columns from the Gretel PII test split, types 5/5 covered PII types correctly with 0/7

false positives on negative columns drawn from real open data; and after deterministic masking, re-running all validators over the output finds zero (0/360) residual PII — residual PII *detectable by our validators*, a circularity we note explicitly: the leak test can only see what the validator tier sees.

4 Evaluation Design

Suite. Five real-error benchmarks (Raha) plus seeded error injection (typo/OCR/case/whitespace) over 15 harvested open-data domains (NYC, Chicago, SF, LA, Seattle, Texas, WA portals; GitHub) ≈ 65 datasets per seed. We aggregate as a *double macro*—mean over error types of mean over datasets, harmonically combined with the domain macro—so no single table or error type dominates:

$$\text{NORTH} = \text{HM} \left(\underbrace{\frac{1}{|T|} \sum_{t \in T} \frac{1}{|D_t|} \sum_{d \in D_t} F_1(d)}_{\text{error-type macro}}, \underbrace{\frac{1}{|G|} \sum_{g \in G} \frac{1}{|D_g|} \sum_{d \in D_g} F_1(d)}_{\text{domain macro}} \right),$$

where T is the set of error types, G the set of data domains, D_t (resp. D_g) the datasets carrying error type t (domain g), and HM the harmonic mean.

Churn-neutral metric. A cell change that is case/whitespace-equivalent to the input but does not restore the gold counts as nothing: not a fix, not a change, not damage. Without this, mass case-rewriting inflates precision (we observed +0.12 NORTH from *removing* case matching before the correction); with it, fixing a case-injected error requires actually acting. We additionally report *damage*—the rate of semantically corrupting clean cells—and an adversarial *abstain slice* whose traps are garbage strings (not single-edit variants of any reference entity; an earlier trap set mis-scored grounding for correctly mapping Boazz→Boaz). We report both repairs of these metric artifacts as evidence that gameability must be tested, not assumed.

Real vs. injected. Injected typos are in-distribution for frequency clustering by construction (the canonical is present and dominant in the column), so we report the real-error and injected slices separately.

Scorer validation. Following GroUSE-style evaluator testing [26], the scorer itself is validated against 30 adversarial known-by-construction cases: a no-op plan must score 0 fixes and 0 damage, an oracle plan exactly 1.0, vandalizing k of m clean cells must score damage k/m at precision 0, pure churn (case/whitespace rewrites that do not restore gold) must count as nothing although a naive scorer would count it, fixes must require actually acting, and silent edits must trip the audit. All 30 pass against the shipped scorer unmodified.

Contamination. The Raha-suite benchmarks have been public on GitHub since 2019 and sit inside every modern base model’s training window; we treat them as potentially contaminated and split our claims accordingly. A verbatim-completion probe makes the concern concrete: prompted with five fields of a gold hospital row, a frontier-class model reproduces **25%** of the held-out cells exactly (30/120 cells over 30 rows, exact-substring match), versus **0%** (0/120) on a date-stamped post-training-cutoff wild harvest under the identical protocol (`eval/contamination_probe.py`). The rate is an upper bound on memorization — some completions are guessable from the given fields — but it is not zero, so results on legacy-public benchmarks (including the all-hospital Table 2, whose zero-shot planners may partially benefit from memorized gold) carry this caveat, while the architecture’s trust claims (zero silent edits, damage accounting, abstention) rest on the date-stamped wild and GitTables slices, where the probe finds nothing to complete.

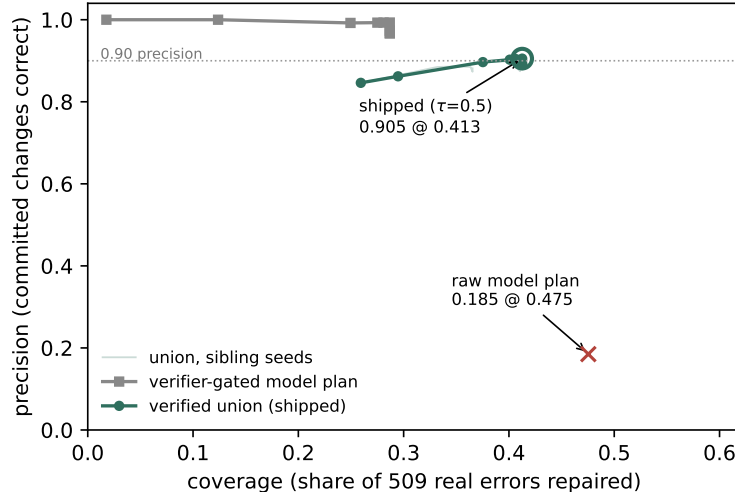


Figure 1: Plan-level precision–coverage on hospital (509 real errors), sweeping the verifier threshold τ . The union planner dominates the raw model plan; the shipped operating point ($\tau=0.5$) is annotated.

5 Results

5.1 Plan-level selective prediction on real errors

On hospital’s 509 real errors, the verifier transforms the fine-tune from unshippable to precise (Figure 1): the raw model plan repairs 0.475 of errors at 0.185 precision; gated at $\tau=0.5$ it reaches 0.993 precision at 0.287 coverage (146 of 147 committed changes correct). The union with the grounded heuristic buys coverage back: **0.905 precision at 0.413 coverage** (232 changes, 210 correct). This turns the system’s promise into a measured sentence: *zero-configuration and zero labels, repair 41% of real errors at ≥ 0.90 precision, with every declined merge surfaced for review*. For context, Baran given oracle error positions and 20 gold-labeled tuples per dataset reaches 0.811 F1 on the same slice (§5.8)—selective prediction does not close a supervised gap, but it makes the zero-label operating point trustworthy, which is the regime our user occupies. Precision is flat (0.89–0.91) for $\tau \in [0.2, 0.8]$, so the operating point is not threshold-brittle, and the result is seed-robust: across three training seeds of the same data recipe the union operating point is 0.891 ± 0.012 precision at 0.396 ± 0.025 coverage (the shipped adapter is the strongest seed), with every seed clearing the 0.70-precision/0.30-coverage bar decisively.

Candidate-constrained planning (negative result). We also tested constraining the planner’s *inputs*: the profiler emits evidence-backed (variant $\rightarrow$ canonical) candidate pairs (frequency dominance, edit similarity, reference membership) and the model may only select among them, with a deterministic check dropping off-candidate mappings to review flags. As a standalone guard it is strong—the raw plan’s precision rises from 0.185 to 0.760 with no verifier at all—but composed with the verifier and union it reaches 0.876 precision at 0.387 coverage, slightly *below* the unconstrained pipeline at the same τ : the candidate cap (top-3 per surface) removes some correct repairs the verifier would have kept, and the two mechanisms gate the same failure class. We ship the verifier and keep candidate constraining available but off by default, reporting this as a measured redundancy rather than a stacked win.

5.2 The 4B fine-tune as one planner instantiation

On frozen synthetic gold, the fine-tuned 4B planner reaches canonicalization micro-F1 0.803 ± 0.009 (95% CI, 3 training seeds) — versus 0.452 for a much larger zero-shot generalist prompted identically and 0.152 for the rule heuristic (best single run 0.901; operation-F1 0.957, JSON validity 0.950). On real hospital typos the synthetic-only fine-tune scores 0.000 repair recall; adding 20% real-derived supervision lifts it to 0.424, and a data-scaling iteration (tripling the real-derived share from three paired benchmarks) reaches 0.475 recall at 0.185 precision—approaching the 0.51 of a frontier-scale zero-shot model. The scaling gain is seed-robust: +0.09 canonicalization F1 over the base mix under identical protocol, with non-overlapping 3-seed confidence intervals. Real, execution-verified pairs are what transfer: the same iteration found frequency-derived and algorithm-cleaned labels both *reduce* quality, consistent with our grounding thesis.

5.3 Grounding vs. clustering

With the errors-are-rare frequency gates now in both paths, grounding and frequency clustering are comparable on hospital alone (repairs-only, churn-neutral: 0.845 precision at 0.257 recall grounded vs 0.871 at 0.293 clustering—hospital’s dominant errors are in-column typos, clustering’s best case). Grounding’s margin appears where references matter: across the five-benchmark real-error macro it reaches 0.174 versus 0.135 for the frequency-clustering ablation (+29%), and it carries the behavioral guarantees below. On the full suite against OpenRefine (Table 1), the result splits cleanly by regime, and we report both. On the *real-error* slice—the regime the tool exists for—grounded cleaning reaches REAL-F1 0.225, $3.9\times$ OpenRefine kNN (0.058) and $5.7\times$ fingerprint (0.039), with seed CIs of ± 0.004 . Provenance: the grounded and OpenRefine rows of Table 1 are regenerated at the current system head (2026-06-12, post-capability, scorer fix in); the dagger rows keep their original capture provenance. The June-10 freeze system measured REAL-F1 0.174 on the same protocol — the +0.05 difference is the measured contribution of the four deterministic capabilities (§3.5) on the real-error slice. On the *injected* slice, fingerprint clustering wins (0.282 vs 0.224) at near-zero damage: our case/whitespace injectors are exactly the perturbations key-collision normalizes away, so this is its home game and we say so. kNN clustering—the method that, like us, attempts typo repair—loses on both slices while incurring the highest damage among baselines (0.096), the no-reference over-merging failure the grounding was built to prevent. The shipped verified-union system’s suite row (REAL-F1 0.142, damage 0.015) shows the grounding wrapper and heuristic union carry entity canonicalization on these datasets — the model’s contribution concentrates on the synthetic regime and hospital repair (§5.1), and the verifier cuts its suite damage to 0.015, $6\times$ below the grounded heuristic’s 0.092 (HEAD damage vs the union row’s freeze-time capture — a disclosed basis mix). Within our own ablations (June-10 freeze basis throughout), removing grounding cedes 22% of real-error F1 (0.135 vs 0.174) and forfeits the behavioral guarantees: perfect abstention on adversarial traps (1.000) versus 0.250 without abstention, and reference-vetoed wrong merges (e.g. `guntxrsvillx`→`huntsville`).

5.4 Generalization to never-seen tables

The freeze-version system above was then pointed at data it had never seen, under three new harnesses (all released with this paper as the WILDCLEAN bundle). **(1) Paired bench:** 42 dirty/gold pairs spanning the Raha suite, SemTab ToughTables, government open-data typo corpora, entity-matching tables, and LLM-cleaning evaluation sets. On the 35 pairs from sources absent from training — a count that coincidentally equals, but is distinct from, the 35 gold-free wild tables of harness (2) below — the post-freeze system scores **macro F1 0.363 at damage 0.0219**. The

Table 1: Wide-suite comparison, 3 injection seeds, churn-neutral metric. NORTH is the double-macro harmonic mean; REAL-F1 is the real-error slice. Regenerated at the current system head (2026-06-12); the June-10 freeze system measured 0.174 REAL-F1 / 0.203 NORTH on the same protocol.

System	NORTH	$\pm 95\%$ CI	REAL-F1	INJ-F1	damage	abstain
Grounded (ours)	0.224	0.004	0.225	0.224	0.092	1.000
OpenRefine fingerprint	0.211	0.000	0.039	0.282	0.001	1.000
OpenRefine kNN	0.122	0.002	0.058	0.148	0.096	1.000
Verified union 4B (shipped) [†]	—	—	0.142	—	0.015	1.000
Baran (oracle det. + 20 labels) [‡]	—	—	0.811	—	0.003	—
Jellyfish-13B (ED+DI) [‡]	—	—	0.074	—	0.027	—

[†]single seed, REAL + typo-injected slice only (GPU cost); other rows are 3-seed means. [‡]real slice only, disclosed protocol asymmetries (§5.8): Baran uses oracle error positions + gold labels; Jellyfish is our detect-then-impute composition with seen-data caveats.

largest single contribution is the regime §3.5(b) unlocks: on five all-unique entity tables where no in-column frequency signal exists, F1 moves from 0.0 to 0.955–0.957 at zero damage. Cross-row voting (§3.5c) is the second: flights—many sources reporting the same flight—goes from 0.044 to 0.164 F1 heuristic-only, and the heuristic hospital path doubles from 0.092 to 0.186. The hospital union gate is invariant under all of this (0.905 at 0.413). **(2) Wild bench:** 35 uncurated in-the-wild tables (open-data portals, GitHub, Kaggle) with no gold; we score seeded inject–recovery on each table’s own data (mean recovery 0.207 over the 34 tables with inject scores; one table has none) plus a behavioral audit: every run yields a valid plan, every changed cell is attributable to a logged operation — **zero silent edits across all 35 tables.** **(3) Trust audit at scale:** 239 GitTables tables, same property — 239/239 valid plans, zero crashes, zero silent edits. The held-out-source generalization metric (train and evaluation drawn from disjoint benchmark sources) remains low in absolute terms (GEN-F1 0.058, variant-recall 0.108, damage 0.036): cleaning unfamiliar tables is far from solved, and we report the number to anchor the next section’s claim about *where* the capability that does exist actually lives.

5.5 Where capability lives: a bounded null for fine-tuning

Every attempt to move never-seen-table performance through the model weights failed; every gain in §5.4 came from deterministic machinery plus the verifier. Five further supervised fine-tunes — adding 109k harvested real-world alias pairs (ToughTables-derived, MusicBrainz search hints, RxNorm, OpenFlights), error-dense episode mixes, and a suspects-contract retrain — left held-out GEN-F1 *statistically bounded*: every retrain’s delta is positive but negligible (mean +0.003), never approaching the pre-registered $\delta=0.05$. “Bounded” is a tested equivalence claim, not an eyeballed one [25]: across the five-retrain series the mean held-out GEN-F1 delta (retrain minus champion) is +0.0028 (90% bootstrap CI [+0.0008, +0.0049], strictly positive; 10,000 resamples, seed 42; per-dataset granularity, $n=15$ over 3 held-out sources $\times$ 5 retrains — per-pair deltas do not exist for the retrain series, so within-retrain deltas are clustered and we add a retrain-level robustness check, $n=5$ macro deltas), and TOST rejects effects beyond the pre-registered SESOI of ± 0.05 ($p = 8.0 \times 10^{-16}$; retrain-level check $p = 8.3 \times 10^{-8}$). Two reconciliations make the claim auditable. First, the basis: the equivalence series is scored against the champion’s absolute GEN-F1 of 0.0146, while the 0.058 of §5.4 is the *shipped system* at the post-freeze HEAD with all deterministic capabilities — the

equivalence series scores each retrain’s model-union path at its own capture time, so the two figures share a metric but not a basis. Second, the SESOI: weight interventions move GEN-F1 by at most 0.005, while the deterministic machinery of §3.5 moved the unseen-pair macro from 0.10 to 0.363 — $\delta=0.05$ sits an order of magnitude above the measured weight effect and well below the machinery effect, which is exactly the boundary the test is meant to police. Mixing harvested pairs into the training blend *diluted* the synthetic skill the executor verifies (a monotonic dilution law across mix ratios). A GRPO pilot using the executor as a verifiable reward (the direction RLVR table work [14] motivates) was negative in all three arms at 4B/LoRA scale: the main arm and a KL-anchored variant degraded plan-format validity, and a random-reward control arm reproduced the same drift, identifying it as an RL artifact rather than signal [15]. We state this as a *bounded* null, not a universal one: at 4B/LoRA scale, under our propose/execute protocol and training budgets, no weight intervention we ran produced measurable movement in never-seen-table repair — profiling visibility, reference grounding, cross-row consensus, convention conservatism, and plan-level verification carry the capability that exists. The bound is explicit: results with full-scale RL infrastructure (execution-verified rewards on multi-GPU RLVR stacks [11, 14]) show task skill moving at the same parameter scale, so our claim is about what SFT-and-pilot-RL buy in this protocol class, not about reinforcement learning in general. A second explicit bound: every weight experiment here uses the Qwen3 family — and the very work we cite to explain the control arm’s drift documents that random-reward GRPO effects are themselves family-sensitive [15] — so the null is stated for Qwen3-class models pending a cross-family replication. Concurrent evaluations corroborate the mechanism from independent directions [12, 13]. The practical corollary is unusual but actionable: a contributor who wants to improve a system like this should write a deterministic capability and gate it with the verifier, not collect more training data.

The null extends to test-time compute — with one instructive exception that *confirms* the architecture claim. Self-consistency *voting* over $N=16$ temperature-0.7 samples (cell-edit-level majority, run through the identical verifier-union pipeline) yields 0.906 precision at 0.454 coverage versus 0.9055 at 0.4519 for matched greedy decoding on the same local runtime — a null at matched precision, the visibility law from the test-time side: voting cannot surface repairs the profile does not expose, and it actively discards verified-recoverable coverage. But pooling *every* mapping from all 16 samples and letting the verifier filter the union gives the best operating point we measure for the 4B: **0.911 precision at 0.483 coverage** (+0.6 points precision, +7.1 points coverage over the shipped gate; two independent runs agree to ± 0.001). The lesson is the paper’s thesis in miniature: sampling helps only as a *candidate generator*; consensus adds nothing the verifier does not already provide — pool candidates, verify, do not vote. Separately, the local capture path itself (Q8 quantization with grammar-constrained decoding) is worth +3.9 points of coverage over the original Modal capture at equal precision.

5.6 Zero-label capability scaling: the verifier harness is planner-agnostic

The negative result bounds what fine-tuning small weights buys; it says nothing about raw capability. To separate the two we dropped zero-shot, $\leq 32B$ open-weights planners — with *no* task training — into the identical hospital pipeline the 4B fine-tune uses: same prompt contract, same verify($\tau=0.5$), same union with the grounded heuristic (Table 2). devstral-small-2-24B and gemma4-31B both reach **0.915 precision at 0.485 coverage** — exceeding the fine-tune’s union point of 0.905 at 0.413 — while nemotron-30B reaches 0.877 at 0.336 with JSON-plan validity 0.4 (validity is part of the measurement: a planner that cannot reliably emit the plan schema loses coverage before capability is measured). gpt-oss-20B is excluded as a serving failure, documented rather than scored as capability: the hosted proxy returned empty content on every planning call

Table 2: Zero-shot ≤ 32 B planners in the identical verify($\tau=0.5$)+union harness, hospital’s 509 real errors. Validity = fraction of planning calls returning schema-valid JSON. Runtime = wall-clock for the planning calls on hosted endpoints (single capture, no seeds; the 4B row is a prior Modal A100 capture with no comparable local figure). Each scaling row is a single capture; the primary evidence is the union coverage delta (+0.07) at matched-or-better precision, not any single cell. Bold marks the best union operating point. gpt-oss-20B excluded (serving failure: empty proxy responses, not measurable capability). The identical devstral/gemma rows are a verified counting coincidence, not a scoring artifact: their applied cell-edit sets share 266 of 270 cells, each commits 4 model-specific repairs (all correct), and the totals coincide (eval/results/scaling_coincidence.json).

Planner	Params	Gated P@C	Union P@C	Validity	Runtime (s)
ScrubData-v6 (Qwen3-4B fine-tune)	4B	0.993 @ 0.287	0.905 @ 0.413	—	—
devstral-small-2 (Mistral)	24B	0.943 @ 0.426	0.915 @ 0.485	1.0	135
nemotron-3-nano (NVIDIA)	30B	1.000 @ 0.138	0.877 @ 0.336	0.4	114
gemma4 (Google)	31B	0.943 @ 0.426	0.915 @ 0.485	1.0	104

despite full-length generation. The arm is multi-family (Mistral, Google, NVIDIA), which addresses the single-family bound of §5.5 for the inference side; the weight-training null itself remains Qwen3-scoped. Disclosure: these models were measured via hosted inference for speed; all are ≤ 32 B open weights and locally deployable in principle. The interpretation we draw is the paper’s sharpest: SFT at 4B does not buy held-out generalization (§5.5), but raw capability at 24–31B does lift the same harness — the verifier/union architecture is the portable contribution, converting any sufficiently capable planner into a trustworthy cleaner.

5.7 Ablations

All ablations are 3-seed means (CIs $\leq \pm 0.003$). Removing abstention costs -0.013 NORTH, raises damage to 0.108 (from 0.104), and collapses trap abstention to 0.250. Removing the ambiguity margin costs -0.006 with $+0.001$ damage. Removing case matching costs -0.002 under the churn-neutral metric (and *gained* $+0.12$ under the uncorrected metric—the artifact). Replacing grounding with frequency clustering gains $+0.020$ NORTH, all of it from the injected slice (§4), while ceding -0.039 real-error F1—the trade the system refuses by design.

5.8 Learned-repair baselines under disclosed protocols

We additionally run two learned-repair baselines on the real-error (Raha) slice, under the identical churn-neutral metric but with honestly disclosed protocol asymmetries. **Baran** [1] is semi-supervised: we run its reference configuration—oracle error positions from the dirty/gold diff plus 20 gold-labeled tuples per dataset (its package default), without the optional Wikipedia-pretrained value models. It reaches REAL-F1 0.811 ± 0.018 (3 label-sampling seeds) at 0.003 damage—an upper bound under a strictly more informed protocol than ours (zero labels, no oracle detection); with oracle positions it can essentially only edit true-error cells, so its near-zero damage is structural. **Jellyfish-13B** [5] publishes per-cell error detection and imputation but no repair task; we compose the two (detect, then impute flagged cells with the attribute masked) — a pipeline of our construction, not theirs. It scores REAL-F1 0.074 at 0.027 damage (single seed, recommended decoding; note hospital is in its instruction-tuning data and flights/rayyan in its published evaluation suite, so these numbers may flatter it). Neither baseline is run on the 56-spec injected suite (computationally and methodologically out of scope for semi-supervised and per-cell-LLM repair); their

Table 3: Per-dataset real-error results (Raha slice), churn-neutral F1 / damage. Grounded is the HEAD deterministic system; OR = OpenRefine reimplementations; Union is the verified union planner ($\tau=0.5$) where a captured model plan exists (movies_1 capture pending); Baran uses oracle error positions + 20 gold labels (mean of 3 label-sampling seeds) and is a supervised reference, not a peer.

Dataset	Grounded (HEAD)	OR fingerprint	OR kNN	Verified union	Baran (oracle+20)
hospital	0.258 / .066	0.000 / .000	0.189 / .083	0.567 / .001	0.827 / .004
beers	0.025 / .005	0.194 / .000	0.086 / .074	0.035 / .001	0.918 / .000
flights	0.127 / .082	0.000 / .000	0.014 / .065	0.035 / .080	1.000 / .000
rayyan	0.000 / .118	0.000 / .001	0.002 / .008	0.000 / .001	0.402 / .010
movies_1	0.714 / .025	0.002 / .018	0.001 / .072	—	0.909 / .001
macro F1	0.225	0.039	0.058	—	0.811

NORTH/INJ-F1 cells in Table 1 are blank by design. The comparison locates our contribution: zero-config systems (ours, OpenRefine) occupy a different protocol class from supervised repair, and the verifier (§5.1) is what makes the zero-config class precise enough to trust, not what closes the labeled gap.

Table 3 breaks the real-error slice down per dataset at HEAD. The verified-union rows are reported with their honest shape: off hospital the union turns ultra-conservative — on rayyan it commits 12 changes at 0.001 damage; on beers it holds precision 0.546 at recall 0.018. The gate’s precision premise transfers as *safety* (union damage stays at 0.001–0.080) but not as coverage; the movies_1 plan capture is pending. This directly answers the co-adaptation concern: hospital is where the model’s learned mappings live, and elsewhere the system abstains rather than guesses.

5.9 A matched label budget separates the supervision regimes

The Baran comparison above is two points (zero labels, twenty labels); the matched-budget curve in Figure 2 measures what each label is worth to each system on the same five-dataset real-error macro. At zero labels Baran — even *retaining* its oracle error positions — repairs nothing (F1 0.000, 3 seeds): its value models have nothing to learn from. ScrubData operates at 0.225 with zero configuration. With labels Baran climbs steeply (0.504 at $k=5$, 0.811 at $k=20$): the two systems occupy complementary supervision regimes, a relationship now measured rather than asserted. ScrubData’s own k -label arm uses the labels *only* to validate and expand the verifier accept set — no retraining, no oracle positions: 0.282 ± 0.023 at $k=5$ and 0.351 ± 0.012 at $k=20$ (3 label-sampling seeds). The disclosed asymmetry stands at every budget: Baran keeps oracle error positions throughout, so the curve is an upper bound in its favor.

5.10 Degenerate baselines and cost-weighted damage

Four degenerate policies pin the metric’s floor and ceiling on the full 42-pair bench (Table 4). No-op and oracle land exactly at 0 and 1; abstain-all is score-identical to no-op because the repair metric is flag-blind by design (abstentions are audited separately); seeded random editing of 5% of cells is vandalism the metric must punish. Since F1 alone under-punishes vandalism, we add a cost-weighted score in the Effective-Reliability style, $\Phi_c = (\text{fixes} - c \cdot \text{damaged}) / \text{errors}$ at $c \in \{1, 5, 10\}$: random editing scores -0.49 to -4.89 , while the shipped system stays positive at $c=1$ (+0.13) — and goes negative at higher c , which is the honest reading: at 10:1 cost asymmetry, only near-zero-damage operating points (the verified union) are defensible.

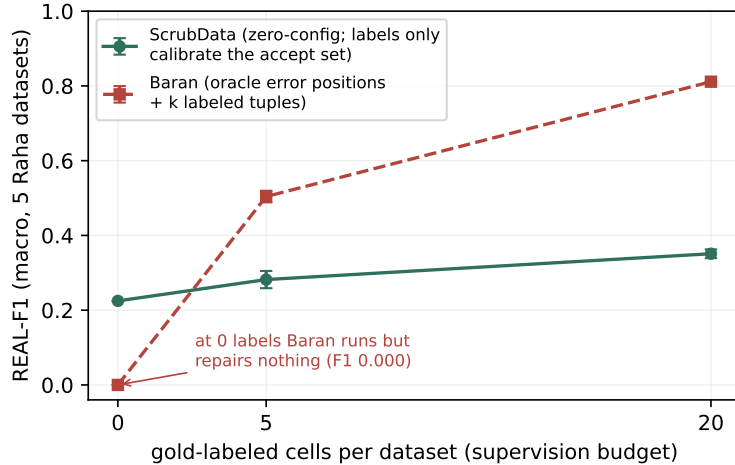


Figure 2: Matched-budget label curve, five-dataset real-error macro F1. At $k=0$ Baran repairs nothing even with oracle error positions retained; ScrubData operates at 0.225 with zero configuration. With labels Baran climbs steeply — complementary supervision regimes, measured. Error bars ($\pm$) are standard deviations over 3 label-sampling seeds; the Baran $k=20$ point reuses the 3-seed baseline run of §5.8.

Table 4: Degenerate policies pin the metric (42 pairs, churn-neutral macro; random-edit: seeded, 5% of cells). Φ_c is micro-summed (fixes $-c \cdot$ damaged) per benchmark error. “Shipped” here is the deterministic grounded path on the 42 pairs (damage 0.023), distinct from the verified-union suite row of Table 1 (damage 0.015).

Policy	F1	P	R	damage	Φ_1	Φ_5	Φ_{10}
no-op	0.000	1.000	0.000	0.000	0.00	0.00	0.00
abstain-all	0.000	1.000	0.000	0.000	0.00	0.00	0.00
random-edit	0.000	0.001	0.001	0.049	-0.49	-2.45	-4.89
oracle	1.000	1.000	1.000	0.000	+1.00	+1.00	+1.00
shipped	0.343	0.576	0.308	0.023	+0.13	-1.37	-3.26

One disclosure: the oracle acceptance check itself surfaced a scorer artifact — 3 cells in 1.79M held the literal string `Nan` (a first name), which parses to float NaN and was unequal to itself — now fixed in `eval/metrics.py` with a regression test; published numbers shift by less than 10^{-4} .

5.11 Calibration of abstention

On a probe of reference-entity typos plus garbage traps, retrieval confidence is a usable selective-prediction signal: AURC 0.120, ECE 0.169 (over-confident; temperature scaling is future work), and precision rises monotonically with threshold—0.899 precision at the default $\tau=0.84$ (coverage 0.669), and $\geq 95\%$ precision at $\tau=0.91$ (coverage 0.206). Figure 3 shows the risk-coverage curve.

6 Limitations

Reference coverage is the recall ceiling: entities absent from the taxonomy abstain by design, which is safe but not helpful; coverage work (larger gazetteers, ROR for organizations) moves recall

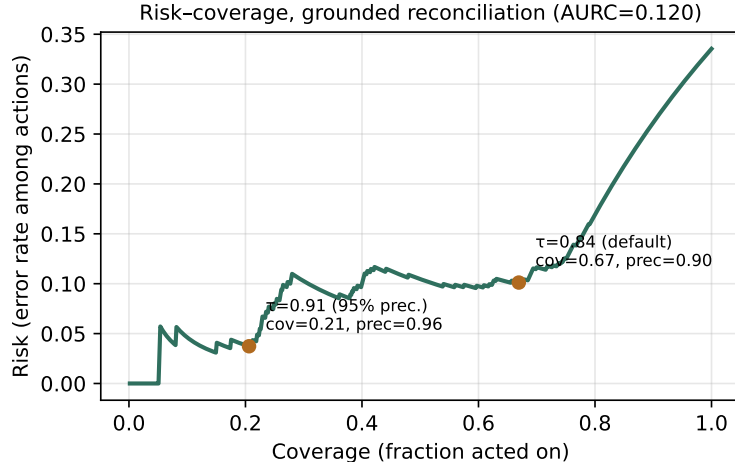


Figure 3: Risk-coverage for grounded city reconciliation (650 probes). Operating points annotated; the confidence supports thresholded abstention.

directly. Our damage metric is convention-tolerant for case and whitespace but still counts alias expansion (NYC→New York) as damage when the gold keeps the alias—a value-level convention question we leave open. The confidence signal is over-confident (ECE 0.169); temperature scaling is future work. The injected half of the suite, while seeded and reproducible, inherits the injector’s error model; we mitigate with the real-error slice and report both. All weight-training experiments (SFT and GRPO) use a single model family (Qwen3), so the negative result of §5.5 is family-scoped until replicated on a second family. PII coverage is English-only, and we make no de-identification guarantee. Finally, the fine-tune headline is reported with multi-seed confidence intervals, but the wide-suite model row is single-seed for cost reasons and scoped as such.

7 Conclusion

A planner/executor decomposition with plan-level selective prediction — the model proposes, a deterministic engine executes, a verifier gates every mapping — turns LLM data cleaning from a trust liability into an auditable system: every change is a named, reversible operation; uncertain actions become review flags rather than silent corruptions; and the evaluation itself is built to resist gaming. The post-freeze program sharpened the architecture into a finding: across five further fine-tunes and a three-arm GRPO pilot, the weights never moved never-seen-table performance — deterministic visibility, grounding, consensus, and verification did, at zero silent edits across 35 wild tables and a 239-table trust audit. The scaling arm completes the picture: the bounded null is about fine-tuning small weights, not about capability — two of three zero-shot 24–31B planners dropped into the unchanged verifier harness exceed the fine-tune’s operating point (§5.6), so the architecture is planner-agnostic: capability gains arrive as better operating points without retraining. The shipped system runs entirely locally on commodity hardware and no data leaves the machine; the scaling-arm planners were measured via hosted endpoints, but all are locally deployable open weights. We believe the recipe—propose/execute decomposition, verification-by-execution, retrieval-grounded outputs, and selective prediction over deterministic capabilities—is a template for deploying small specialized models on other structured tasks.

Reproducibility

The model weights are public: <https://huggingface.co/ricalanis/scrubdata-qwen3-4b-v6-q8>. Code, evaluation suite, and result artifacts are released at the project repository, <https://github.com/ricalanis/scrubdata-hackathon> (public upon publication, available to reviewers from the initial submission). The WILDCLEAN bundle — redistributable dirty/gold pairs, the GitTables audit slice, open vocabularies, result JSONs, and license-gated loaders for the non-redistributable pairs — is a public Hugging Face dataset (<https://huggingface.co/datasets/ricalanis/wildclean>). The shipped product planner is the identical code path measured here (`scrubdata/active.py`).

Release integrity. Our own reproducibility QA discovered that the published Q8_0 GGUF was corrupted by an export bug (the export declared a wrong end-of-generation token id inside the Qwen3 vocabulary, degenerating into tool-call loops on all runtimes; a base-model control isolated the fault to the export, not the adapter). It has been re-exported from the v6 adapter and replaced under the same filename, with both sha256 checksums recorded in the model card’s Integrity section. Third-party reproduction of the model-path numbers additionally requires constrained decoding on long prompts — `format=json` under Ollama, or `suppress_tokens=[151657,151658]` under transformers — which is now documented in the model card and `notebooks/Modelfile`.

Setup. Clone the repository and run `uv sync` (Python 3.12; `uv` resolves the pinned environment). The non-redistributable benchmark pairs materialize from their original sources with the WILDCLEAN `loaders.py`. Model-path results additionally need the released Q8_0 GGUF served by a local Ollama (`SCRUBDATA_MODEL`); every deterministic-path number runs with no model at all. Baran runs in the separate pinned environment documented at the top of `eval/run_baran.py`; Jellyfish-13B runs remotely via Modal.

One command per reported number (all from the repository root, at the released revision):

Reported result	Command
Wide-suite comparison (Table 1)	<code>python -m eval.run_real_multi --out eval/results</code>
Precision-coverage curve + gate (Figure 1, §5.1)	<code>python -m eval.precision_curve</code>
Ablations	<code>--plan eval/results/v6_hospital_raw_plan.json --union</code> <code>python -m eval.ablations</code>
Calibration (Figure 3)	<code>python -m eval.calibration</code>
PII leak test	<code>python -m eval.pii_leak</code>
Baran baseline	<code>python eval/run_baran.py</code> , then <code>python -m eval.baselines_learned --score-baran</code>
Jellyfish baseline	<code>modal run scripts/modal_jellyfish.py</code>
Paired bench (§5.4)	<code>python -m eval.paired_bench</code>
Wild bench (§5.4)	<code>python -m eval.wild_bench</code>
GitTables trust audit (§5.4)	<code>python -m eval.gittables_audit</code>
Held-out-source generalization	<code>python -m eval.generalization</code>
Scorer validation (§4)	<code>python -m pytest tests/test_wildclean_scorer.py</code>
Degenerate baselines (Table 4)	<code>python -m eval.degenerate</code>
TOST equivalence (§5.5)	<code>python -m eval.equivalence</code>
Label curve (Figure 2)	<code>python -m eval.label_curve</code>
Per-dataset table (Table 3)	<code>python -m eval.raha_table</code>
Self-consistency null (§5.5)	<code>python -m eval.sc_rerank --model scrubdata-ft --n 16</code>
Scaling arm (Table 2)	<code>python -m eval.scaling_arm</code>

References

- [1] M. Mahdavi, Z. Abedjan, R. Castro Fernandez, S. Madden, M. Ouzzani, M. Stonebraker, N. Tang. Raha: A Configuration-Free Error Detection System. SIGMOD 2019; M. Mahdavi, Z. Abedjan. Baran: Effective Error Correction via a Unified Context Representation and Transfer Learning. PVLDB 13(11):1948–1961, 2020.
- [2] T. Rekatsinas, X. Chu, I. F. Ilyas, C. Ré. HoloClean: Holistic Data Repairs with Probabilistic Inference. PVLDB 10(11), 2017. arXiv:1702.00820.
- [3] J. Peng, D. Shen, N. Tang, T. Liu, Y. Kou, T. Nie, H. Cui, G. Yu. Self-Supervised and Interpretable Data Cleaning with Sequence Generative Adversarial Networks (GARF). PVLDB 16(3):433–446, 2022.
- [4] A. Narayan, I. Chami, L. Orr, S. Arora, C. Ré. Can Foundation Models Wrangle Your Data? PVLDB 16(4):738–746, 2022. arXiv:2205.09911.
- [5] H. Zhang, Y. Dong, C. Xiao, M. Oyamada. Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing. EMNLP 2024. arXiv:2312.01678.
- [6] S. Zhang, Z. Huang, E. Wu. Data Cleaning Using Large Language Models (Cocoon). arXiv:2410.15547, 2024 (preprint; no published reproduction).
- [7] W. Ni, K. Zhang, X. Miao, X. Zhao, Y. Wu, Y. Wang, J. Yin. ZeroED: Hybrid Zero-Shot Error Detection Through Large Language Model Reasoning. ICDE 2025. arXiv:2504.05345.
- [8] M. Wang, J. Wang, Q. Liu, X. Xu, Z. Xing, L. Zhu, W. Zhang. Ensembling LLM-Induced Decision Trees for Explainable and Robust Error Detection. arXiv:2512.07246, 2025 (preprint).
- [9] Q. Chen, Y. He, R. C.-W. Wong, W. Cui, S. Ge, H. Zhang, D. Zhang, S. Chaudhuri. Auto-Test: Learning Semantic-Domain Constraints for Unsupervised Error Detection in Tables. SIGMOD 2025. arXiv:2504.10762.
- [10] M. Yan, Y. Wang, Y. Wang, X. Miao, J. Li. GIDCL: A Graph-Enhanced Interpretable Data Cleaning Framework with Large Language Models. Proc. ACM Manag. Data 2(6), Article 236, 2024 (SIGMOD).
- [11] B. Chi, Y. Xie, M. Wu, J. Yang, J. Jiang, Z. Li, et al. Spreadsheet-RL: Advancing Large Language Model Agents on Realistic Spreadsheet Tasks via Reinforcement Learning. arXiv:2605.22642, 2026.
- [12] A. Dutta, H. Nigam, H. Hasanbeig, A. Radhakrishna, S. Gulwani. An Empirical Investigation of Robustness in Large Language Models under Tabular Distortions. arXiv:2601.05009, 2026.
- [13] C. Parmar, A. Mehta, H. Wu, J. Ramamurthy, S. Medhekar. When Helping Hurts and How to Fix It: Multi-Agent Debate for Data Cleaning. arXiv:2606.02866, 2026.
- [14] Z. Yang, L. Chen, A. Cohan, Y. Zhao. Table-R1: Inference-Time Scaling for Table Reasoning. EMNLP 2025. arXiv:2505.23621.
- [15] R. Shao, S. S. Li, R. Xin, S. Geng, Y. Wang, et al. Spurious Rewards: Rethinking Training Signals in RLVR. arXiv:2506.10947, 2025.

- [16] P. Li, Y. He, D. Yashar, W. Cui, S. Ge, H. Zhang, D. Rifinski Fainman, D. Zhang, S. Chaudhuri. Table-GPT: Table Fine-tuned GPT for Diverse Table Tasks. *Proc. ACM Manag. Data* 2(3), Article 176, 2024 (SIGMOD). arXiv:2310.09263.
- [17] Z. A. Naeem, M. S. Ahmad, M. Eltabakh, M. Ouzzani, N. Tang. RetClean: Retrieval-Based Data Cleaning Using LLMs and Data Lakes. *PVLDB* 17(12), 2024 (demo). arXiv:2303.16909.
- [18] X. Deng, H. Sun, A. Lees, Y. Wu, C. Yu. TURL: Table Understanding through Representation Learning. *PVLDB* 14(3):307–319, 2021. arXiv:2006.14806.
- [19] T. Zhang, X. Yue, Y. Li, H. Sun. TableLlama: Towards Open Large Generalist Models for Tables. *NAACL* 2024. arXiv:2311.09206.
- [20] F. Belotti, F. Dadda, M. Cremaschi, R. Avogadro, M. Palmonari. Evaluating LLMs on Entity Disambiguation in Tables. arXiv:2408.06423, 2024 (preprint).
- [21] L. L. Wei, G. Xiao, M. Balazinska. RACOON: An LLM-based Framework for Retrieval-Augmented Column Type Annotation with a Knowledge Graph. arXiv:2409.14556, 2024 (preprint).
- [22] P. Nguyen, N. Kertkeidkachorn, R. Ichise, H. Takeda. MTab: Matching Tabular Data to Knowledge Graph using Probability Models. *SemTab/ISWC* 2019. arXiv:1910.00246.
- [23] R. El-Yaniv, Y. Wiener. On the Foundations of Noise-free Selective Classification. *JMLR* 11:1605–1641, 2010; Y. Geifman, R. El-Yaniv. Selective Classification for Deep Neural Networks. *NeurIPS* 2017.
- [24] M. Panahi. OpenMed NER: Open-Source, Domain-Adapted State-of-the-Art Transformers for Biomedical NER Across 12 Public Datasets. arXiv:2508.01630, 2025 (preprint).
- [25] D. Lakens. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science* 8(4):355–362, 2017.
- [26] S. Muller, A. Loison, B. Omrani, G. Viaud. GroUSE: A Benchmark to Evaluate Evaluators in Grounded Question Answering. *COLING* 2025. arXiv:2409.06595.